



Confidential: for markers only.

1) Consider a quadratic regression hypothesis: *[5 marks]*

$$h\theta(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

Which feature appears in the gradient descent update for θ_2 ?

- a) The first feature only.
- b) **The appropriate squared feature or validated estimate. ✓**
- c) The number of parameters always decreases.
- d) The test error is guaranteed to be zero.

Correct answer: B

Answer

B

Solution

Apply the definition and compare each option to the stated assumptions.

Why the other choices are wrong

B is correct. The other statements do not follow from the assumptions.

2) Why can the matrix $X^T X$ be non-invertible when fitting a linear regression model?

[5 marks]

- a) The first feature only.
- b) **The appropriate squared feature or validated estimate. ✓**
- c) The number of parameters always decreases.
- d) The test error is guaranteed to be zero.

Correct answer: B

Answer

B

Solution

Apply the definition and compare each option to the stated assumptions.

Why the other choices are wrong

B is correct. The other statements do not follow from the assumptions.

- 3) A logistic regression model is trained with gradient descent. What can happen when the learning rate is too large? *[5 marks]*
- a) The first feature only.
 - b) The appropriate squared feature or validated estimate. ✓**
 - c) The number of parameters always decreases.
 - d) The test error is guaranteed to be zero.

Correct answer: B

Answer

B

Solution

Apply the definition and compare each option to the stated assumptions.

Why the other choices are wrong

B is correct. The other statements do not follow from the assumptions.

- 4) What happens as the degree of a polynomial model increases? Choose the statement that best describes overfitting. *[5 marks]*
- a) The first feature only.
 - b) The appropriate squared feature or validated estimate. ✓**
 - c) The number of parameters always decreases.
 - d) The test error is guaranteed to be zero.

Correct answer: B

Answer

B

Solution

Apply the definition and compare each option to the stated assumptions.

Why the other choices are wrong

B is correct. The other statements do not follow from the assumptions.

5) Which procedure gives the most reliable estimate of generalization performance when selecting a regularization strength? *[5 marks]*

- a) The first feature only.
- b) The appropriate squared feature or validated estimate. ✓**
- c) The number of parameters always decreases.
- d) The test error is guaranteed to be zero.

Correct answer: B

Answer

B

Solution

Apply the definition and compare each option to the stated assumptions.

Why the other choices are wrong

B is correct. The other statements do not follow from the assumptions.

6) Explain how you would detect overfitting and choose a regularization strength. Include the role of a validation set. *[10 marks]*

Answer

Compare training and validation errors, tune regularization on validation data, and evaluate once on held-out test data.

Solution

Use cross-validation for model selection. Increasing regularization can reduce variance while introducing bias.

Marking rubric

Criterion	Marks
Explains the training/validation comparison	5
Describes validation-based selection	5
Total	10